Precision quantifies the repeated-sampling spread of the bridge estimator $\hat{\theta}_S$ as a function of subgroup size N. Panel D shows the 95% confidence interval width under both resampling designs from Panel C:

$$\text{Precision} \;=\; \text{CI width}_{95}(\hat{\theta}_S) \;=\; 3.92 \cdot \text{SE}(\hat{\theta}_S).$$

Two curves appear:

- **Paired** (solid): the optimistic bound from bootstrap subsampling with linked indices. At $N = 500$: CI width ≈ 3.9 SGPc points.
- **Independent** (dashed): the operational bound matching NAEP/TIMSS-style cross-sectional designs. At $N = 500$: CI width ≈ 7.1 SGPc points.

Precision improves sharply from small N into the mid-range (200–800), then flattens. The amber dotted line at 5 SGPc points marks a practical threshold: below this width, subgroup mean *SGPc* becomes operationally useful for reporting and comparison.

The gap between the two curves is the *design effect*: the additional uncertainty from not having linked cohorts. This gap is largest at small N and shrinks (absolutely) as N grows, but the *ratio* remains roughly constant.[4]